

88/100

HW #1

1)

$$a) \frac{dy}{dx} = f(x)g(y) \Rightarrow \int \frac{dy}{g(y)} = \int f(x) dx \Rightarrow \ln|G(y)| = F(x) + C$$

$$G(y) = e^{F(x) + C}$$

$$b) \frac{dy}{dx} + P(x)y = Q(x)$$

$$\mu = e^{\int P(x) dx}$$

$$\mu \left(\frac{dy}{dx} + P(x)y \right) = Q(x)\mu \Rightarrow \frac{d}{dx} \left(e^{\int P(x) dx} y \right) = Q(x) e^{\int P(x) dx}$$

$$\int \frac{d}{dx} \left(e^{\int P(x) dx} y \right) dx = \int Q(x) e^{\int P(x) dx} dx$$

$$e^{\int P(x) dx} y = \int Q(x) e^{\int P(x) dx} dx + C$$

$$y = \left(\int Q(x) e^{\int P(x) dx} dx \right) e^{-\int P(x) dx} + C$$

$$C e^{-\int P(x) dx}$$

(-1)

$$c) y'' + b^2 y = 0 \Rightarrow m^2 + b^2 = 0 \Rightarrow m_1 = bi \quad m_2 = -bi$$

$$y = c_1 \cos(bx) + c_2 \sin(bx)$$

$$d) y'' - b^2 y = 0 \Rightarrow m^2 - b^2 = 0 \Rightarrow m_1 = b \quad m_2 = -b$$

$$y = c_1 e^{bx} + c_2 e^{-bx}$$

$$e) y'' = b_0 + b_1 x + b_2 x^2 \Rightarrow \int \frac{d^2 y}{dx^2} = b_0 + b_1 x + b_2 x^2$$

$$\int \frac{dy}{dx} = b_0 x + \frac{b_1 x^2}{2} + \frac{b_2 x^3}{3} + C_1 \Rightarrow y = \frac{b_0 x^2}{2} + \frac{b_1 x^3}{6} + \frac{b_2 x^4}{12} + C_1 x + C_2$$

$$f) \frac{1}{x} \frac{d}{dx} \left(x \frac{dy}{dx} \right) = b_0 + b_1 x + b_2 x^2 \Rightarrow \frac{d}{dx} \left(x \frac{dy}{dx} \right) = b_0 x + b_1 x^2 + b_2 x^3$$

$$\text{Integrate RHS} \quad x \frac{dy}{dx} = (b_0 x + b_1 x^2 + b_2 x^3) dx \Rightarrow \frac{dy}{dx} = (b_0 + b_1 x + b_2 x^2) dx$$

$$y = b_0 x + \frac{1}{2} b_1 x^2 + \frac{1}{3} b_2 x^3 + C$$

(-3)

$$q) \frac{1}{x^2} \frac{\partial}{\partial x} \left(x^2 \frac{\partial y}{\partial x} \right) + b^2 y = 0, \quad \mu(x) = \frac{u(x)}{x} \quad \frac{\partial u}{\partial x} = -\frac{u(x)}{x^2} + \frac{u'(x)}{x} \quad (1) \frac{1}{x}$$

$$\frac{1}{x^2} \frac{\partial}{\partial x} (x u'(x) - u(x)) + \frac{b^2 u(x)}{x} = 0 \Rightarrow \frac{1}{x^2} (\cancel{u'(x)} + x u''(x) - \cancel{u(x)}) +$$

$$\frac{1}{x^2} (x u''(x)) + \frac{b^2 u(x)}{x} = 0 \Rightarrow \frac{u''(x)}{x} + \frac{b^2 u(x)}{x} = 0$$

$$\frac{m^2}{x} + \frac{b^2}{x} = 0 \quad \frac{m^2}{x} = -\frac{b^2}{x} \Rightarrow m = \pm bi$$

$$u = e^x [C_1 \cos(bx) + C_2 \sin(bx)] = C_1 \cos(bx) + C_2 \sin(bx)$$

$$u = \frac{C_1 \cos(bx) + C_2 \sin(bx)}{x}$$

h) ← missing/incomplete
(-8)

$$i) y'' + b_1 y' + b_2 y = 0 \Rightarrow m^2 + b_1 m + b_2 = 0$$

$$m_1 = \frac{-b_1 + \sqrt{b_1^2 - 4b_2}}{2}$$

$$m_2 = \frac{-b_1 - \sqrt{b_1^2 - 4b_2}}{2}$$

Case I: Distinct Roots:

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

Case II: Repeated Roots (Real):

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x}$$

Case III: Conjugate Complex Root:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad \theta = ? \quad (-1)$$

$$j) y'' + 2xy' = 0$$

$$\frac{\partial u}{\partial x} + 2xu'(x) = 0$$

$$\frac{\partial}{\partial x} (e^{x^2} u(x)) = 0$$

$$\frac{\partial}{\partial x} (e^{x^2} \frac{\partial u}{\partial x}) = 0$$

$$\frac{\partial u}{\partial x} = u(x)$$

$$I = e^{\int 2x dx} = e^{x^2}$$

$$e^{x^2} \frac{\partial u}{\partial x} = C_1$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial x}$$

$$e^{x^2} \frac{\partial u}{\partial x} + 2x e^{x^2} u(x) = 0$$

$$y = C_1 \int_0^x e^{-x^2} dx + C_2$$

$$k) \frac{d}{dx} \left[f(y) \frac{dy}{dx} \right] = 0 \Rightarrow f(y) \frac{dy}{dx} = C_1$$

$$F(y) = C_1 x + C_2$$

$$1) \frac{1}{x^2} \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) = b_0 + b_1 x + b_2 x^2 \Rightarrow \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) = b_0 x^2 + b_1 x^3 + b_2 x^4$$

$$x^2 \frac{dy}{dx} = \int [b_0 x^2 + b_1 x^3 + b_2 x^4] dx = \frac{1}{3} b_0 x^3 + \frac{1}{4} b_1 x^4 + \frac{1}{5} b_2 x^5 + C_1$$

$$\int dy = \int \left[\frac{1}{3} b_0 x + \frac{1}{4} b_1 x^2 + \frac{1}{5} b_2 x^3 + \frac{C_1}{x^2} \right] dx$$

$$y = \frac{1}{6} b_0 x^2 + \frac{1}{12} b_1 x^3 + \frac{1}{20} b_2 x^4 - \frac{C_1}{x} + C_2$$